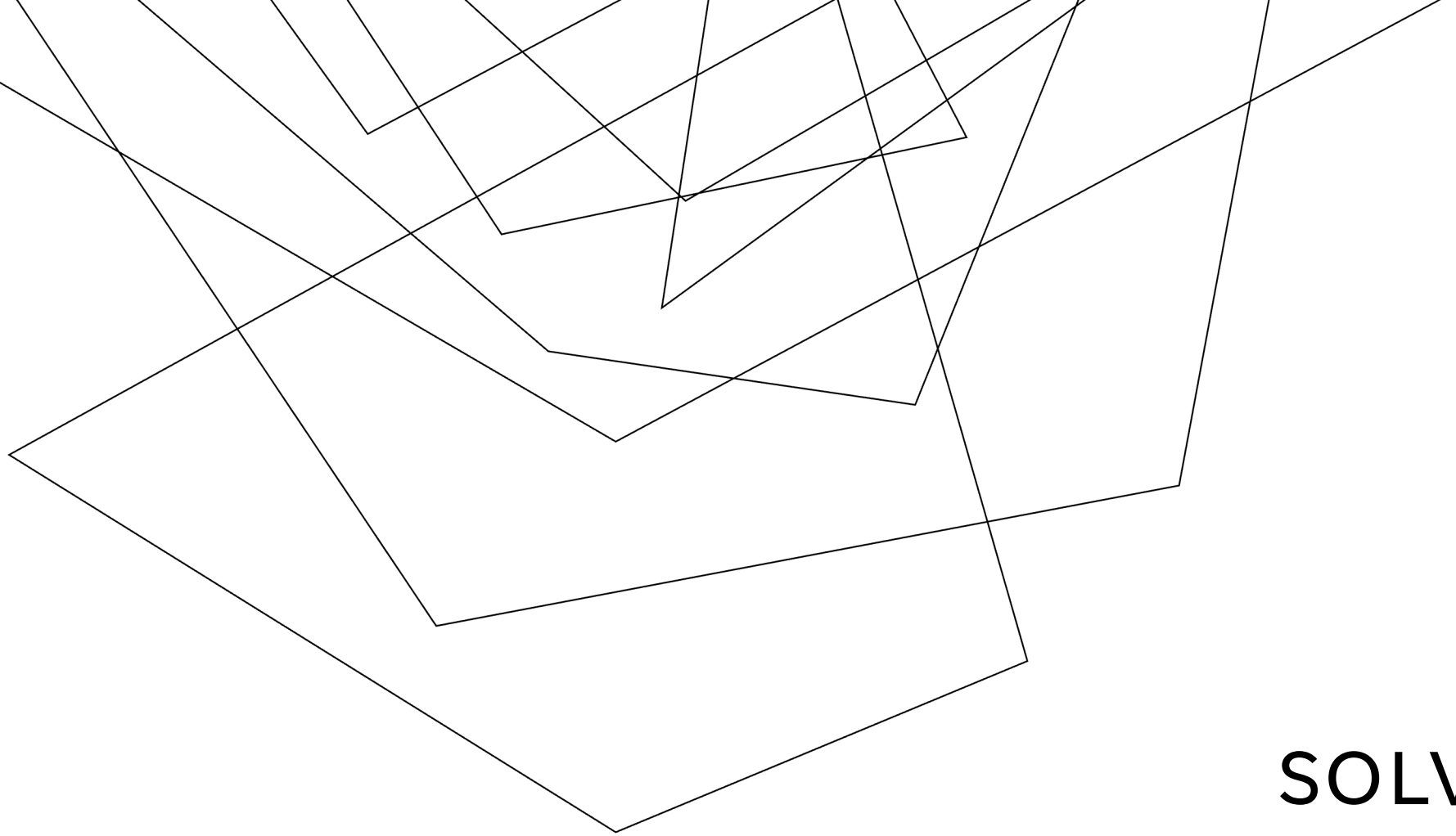
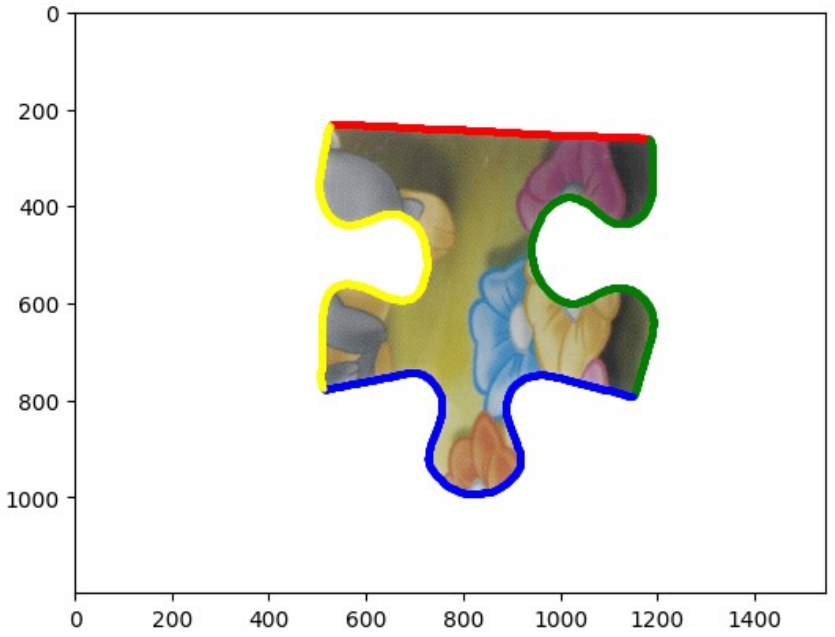


SOLVER

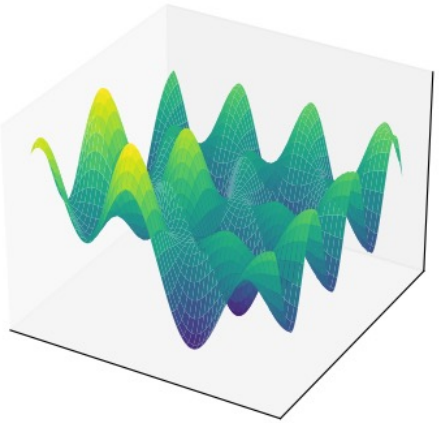
ROBOTIK PROJEKTSTUDIUM
PROF DR. FISCHER - SS24

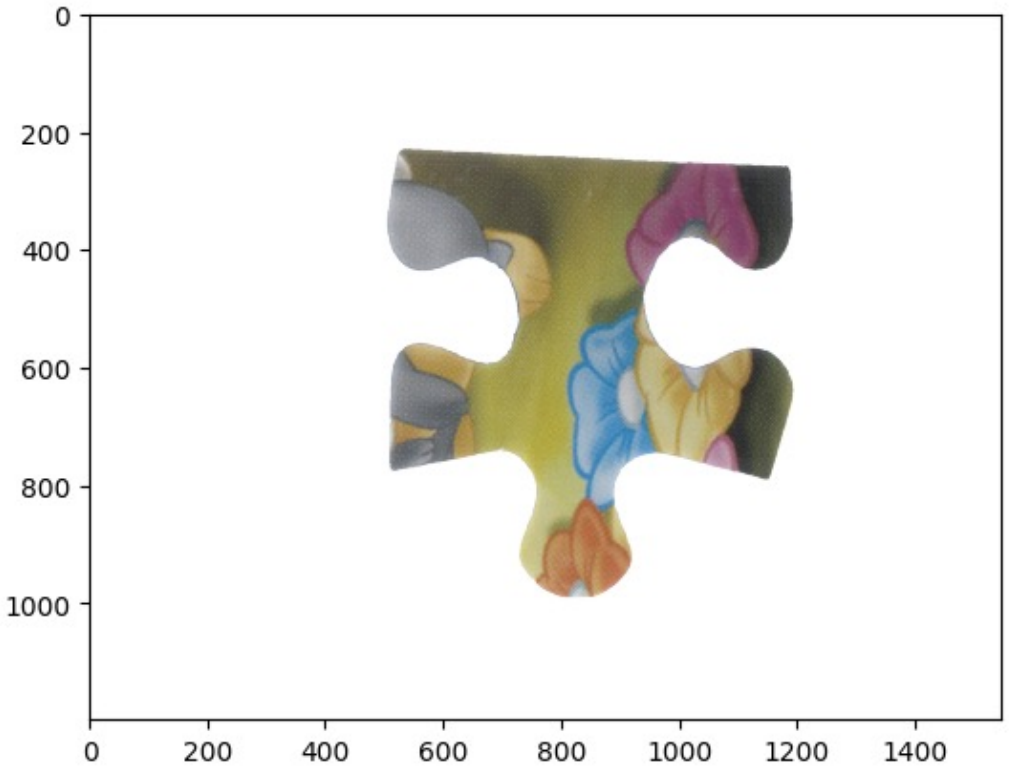
GLIEDERUNG

1. Preprocessing: Edge Extraction
2. Preprocessing: Farb Matching
3. Solving: Simulated Annealing





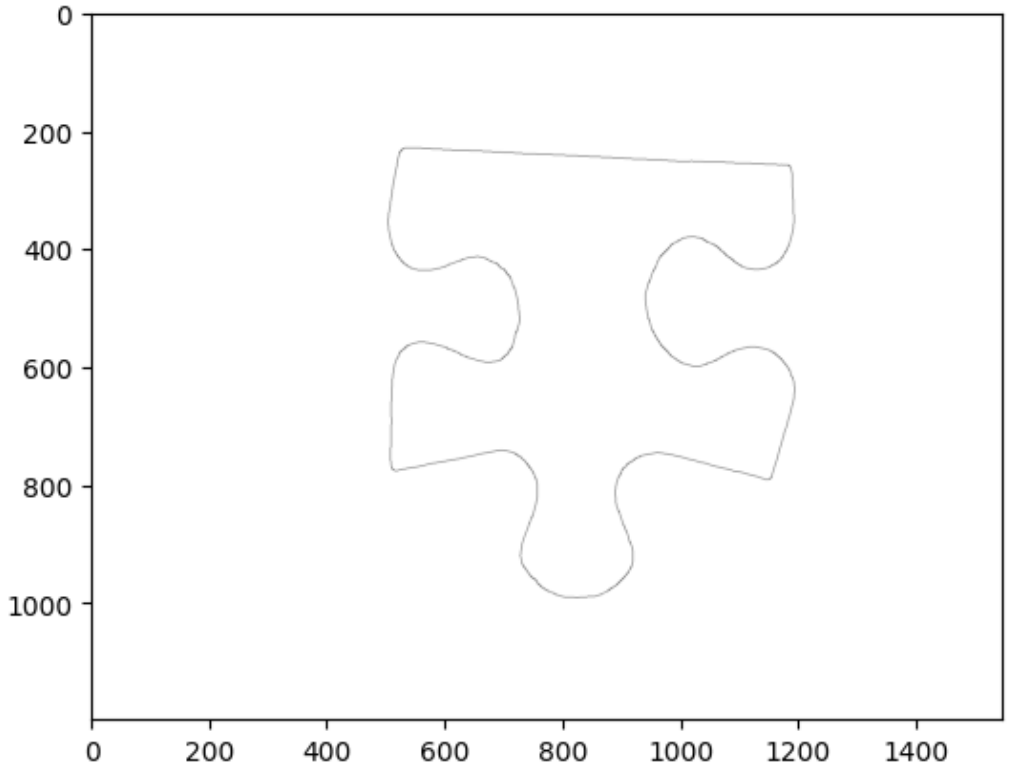




EDGE EXTRACTION

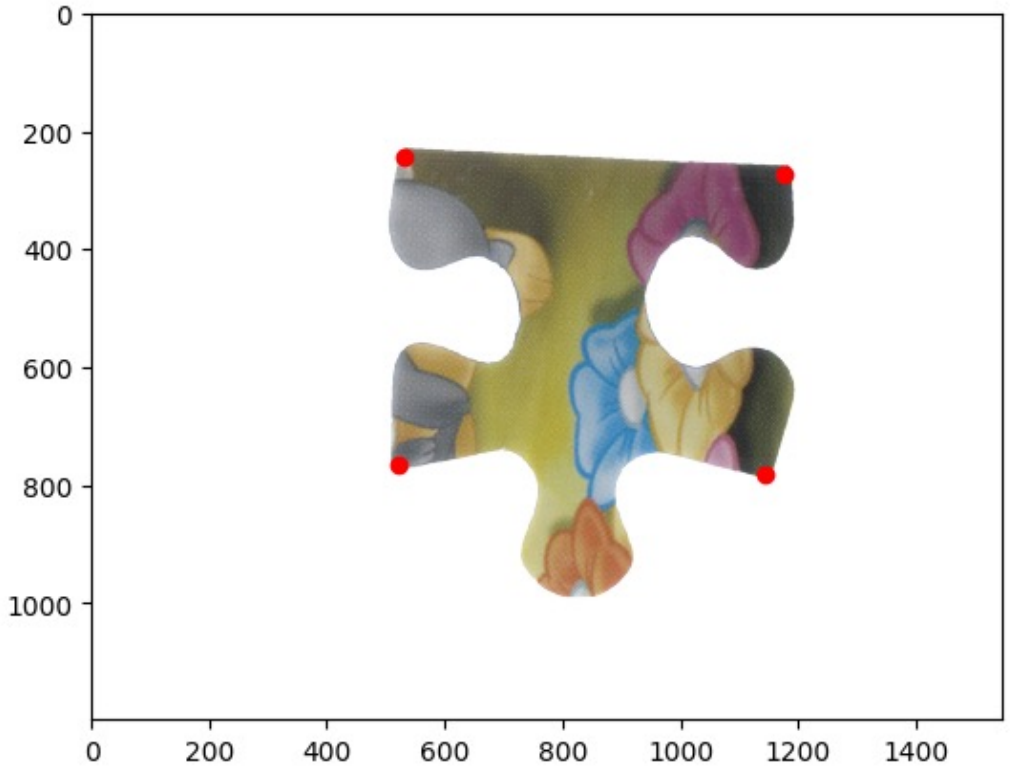
EDGE DETECTION

- Mit OpenCV wird aus dem PNG anhand des Alpha Kanals die Kante extrahiert
- Cv2.Canny()



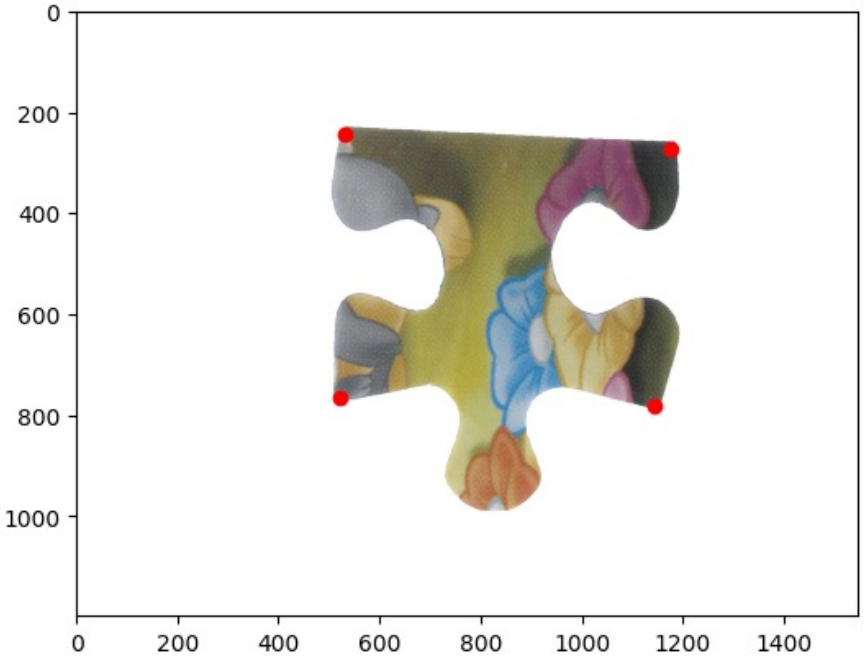
GOOD FEATURES TO TRACK

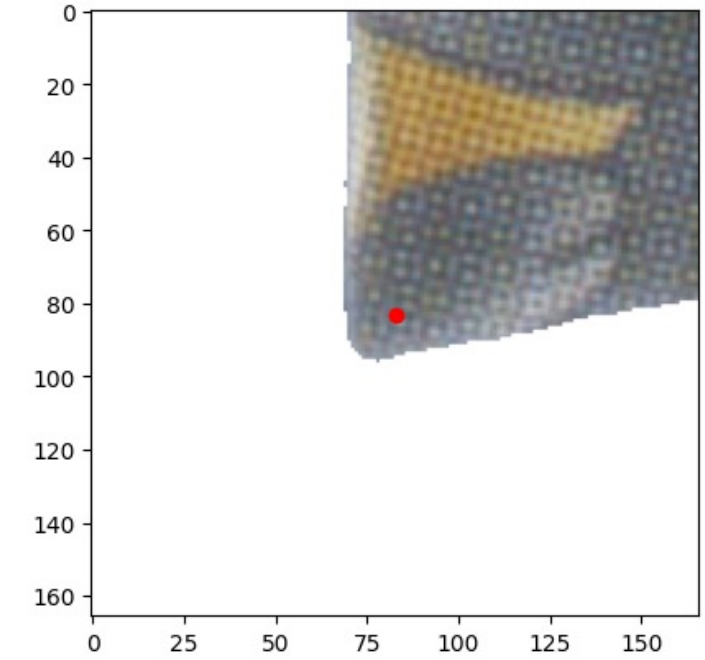
- OpenCV.goodFeaturesToTrack(alpha-Kanal)



GOOD FEATURES TO TRACK

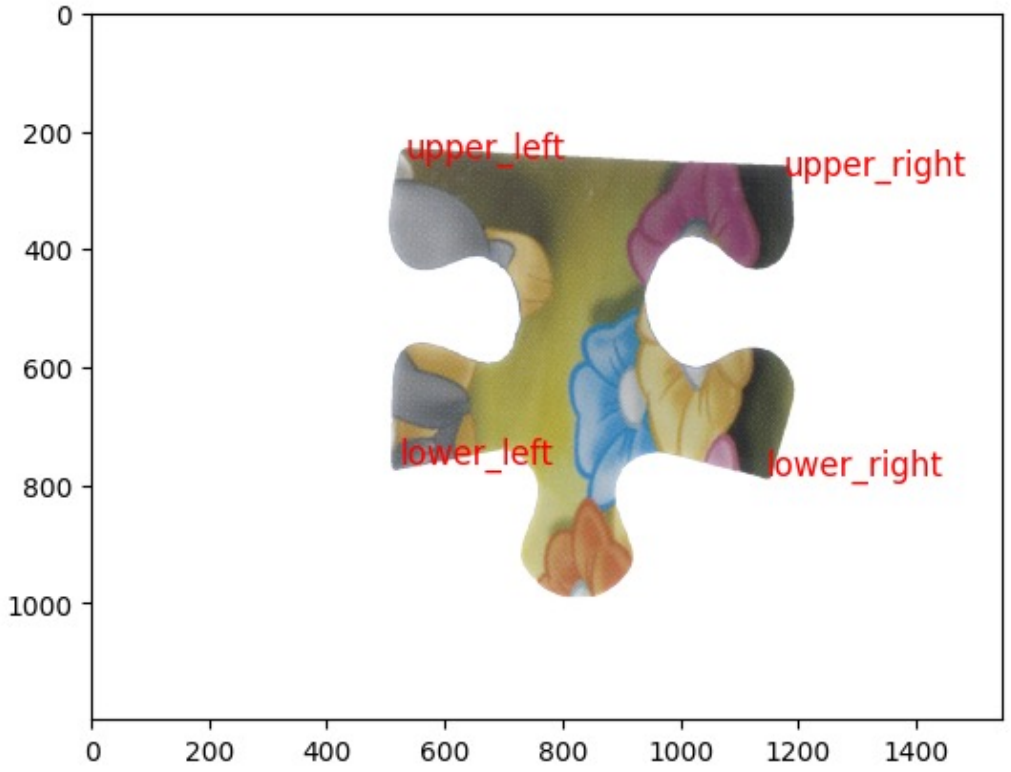
- OpenCV.goodFeaturesToTrack(alpha Kanal)





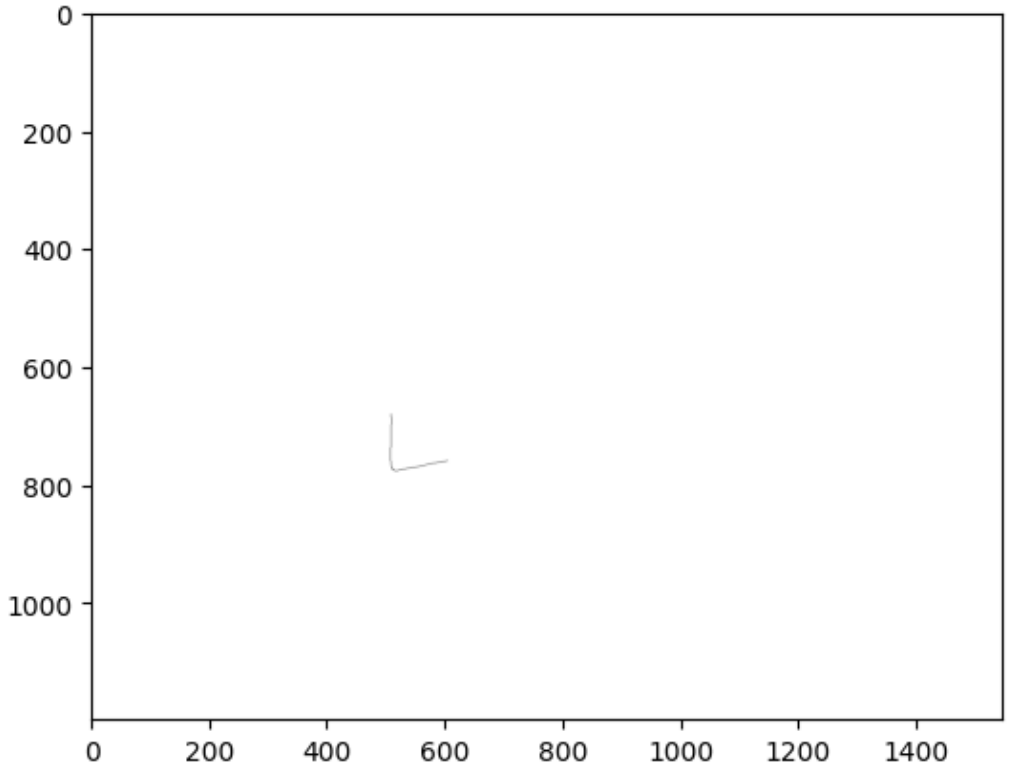
ECKEN LABELN

- Ecken im Uhrzeigersinn definiert



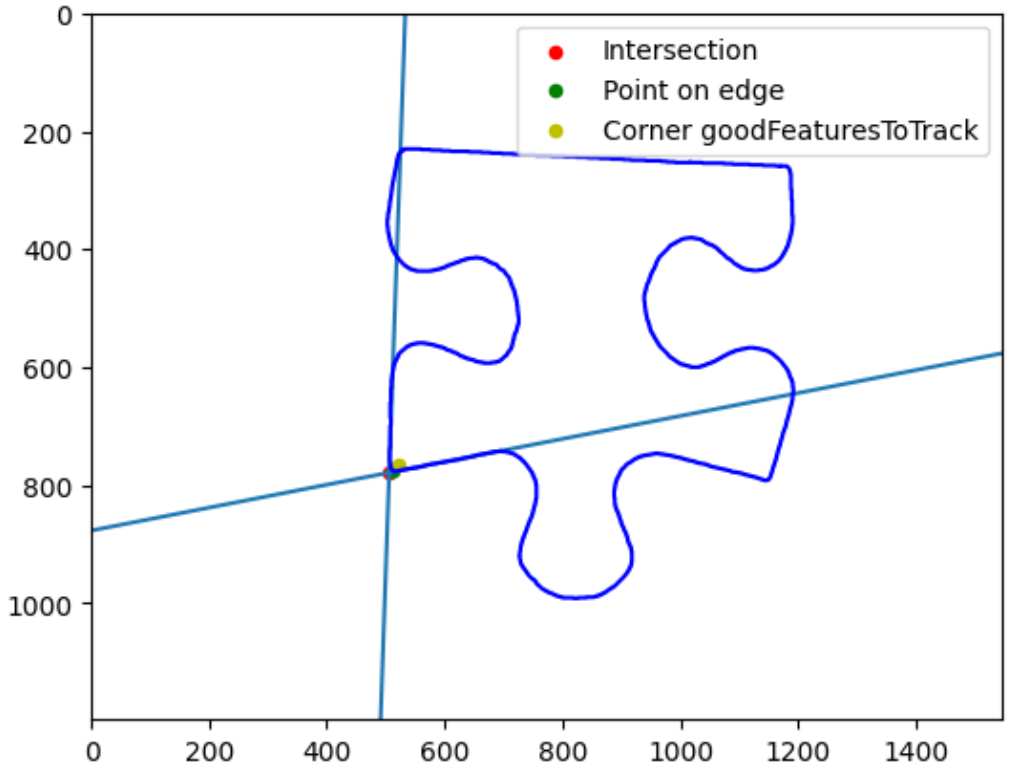
HOUGH TRANSFORMATION

- Hough Transformation erkennt präzise lokal die Kanten
- Regionen anhand der goodFeaturesToTrack Ecken



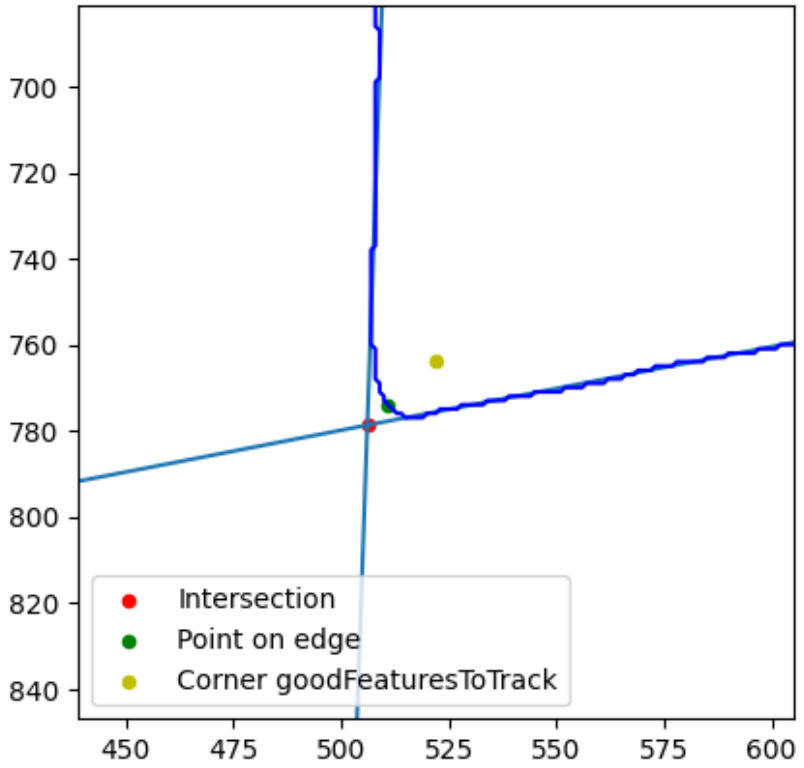
HOUGH TRANSFORMATION

- Erzeugt Geraden auf dem Kantenbild
- Die Schnittpunkte der Geraden werden für die finalen Ecken benutzt



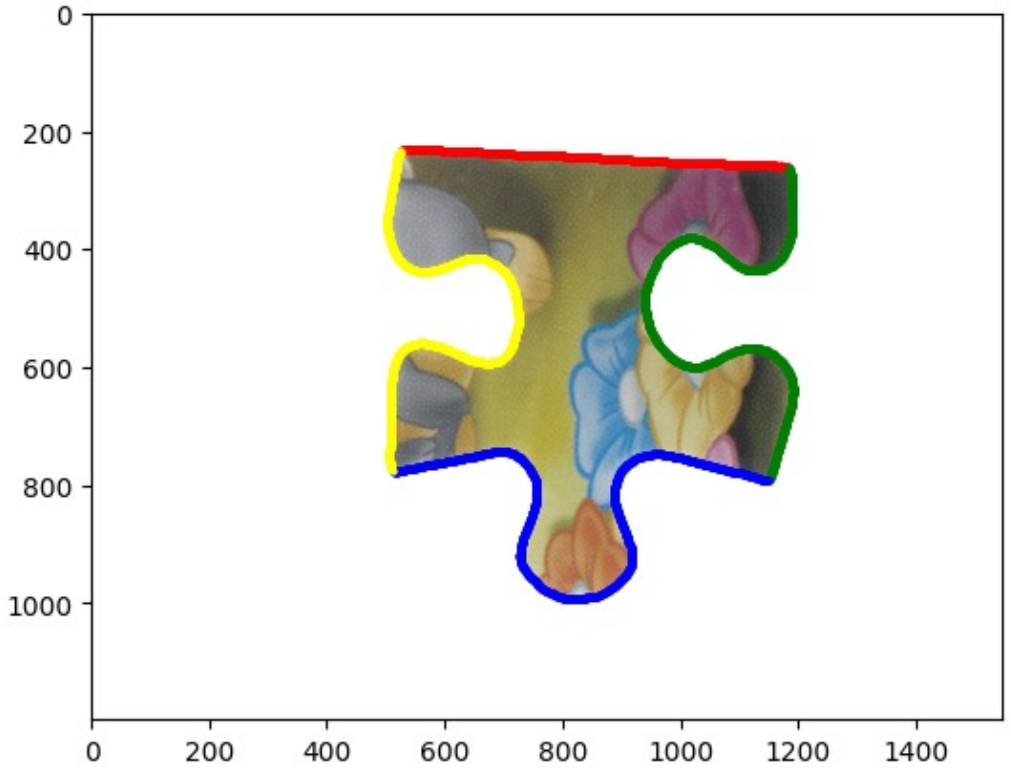
HOUGH TRANSFORMATION

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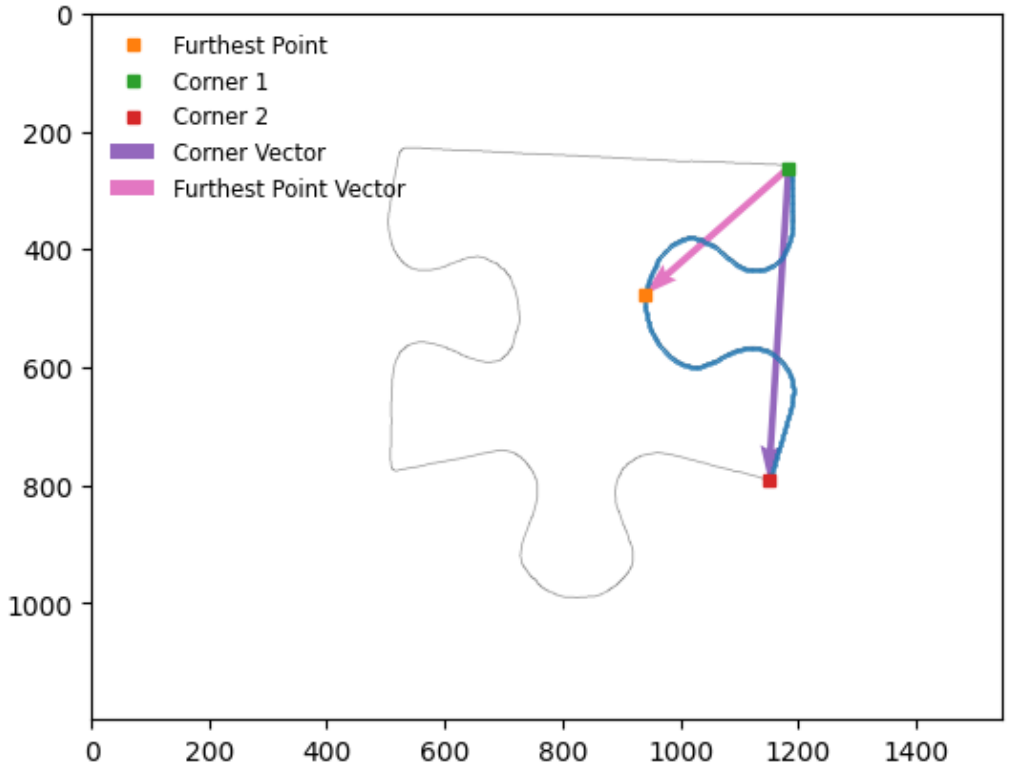
PUZZLE TEIL

- Kante wird an Ecken geteilt
- Ebenso wie Ecken, im Uhrzeigersinn definiert



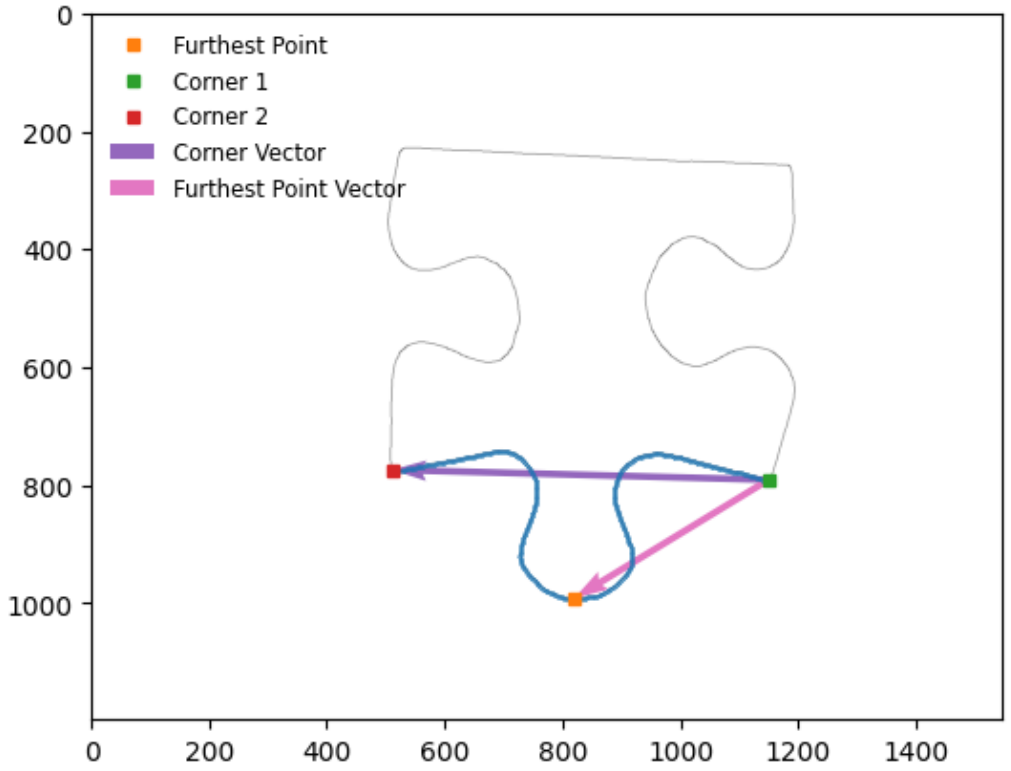
KANTEN TYP

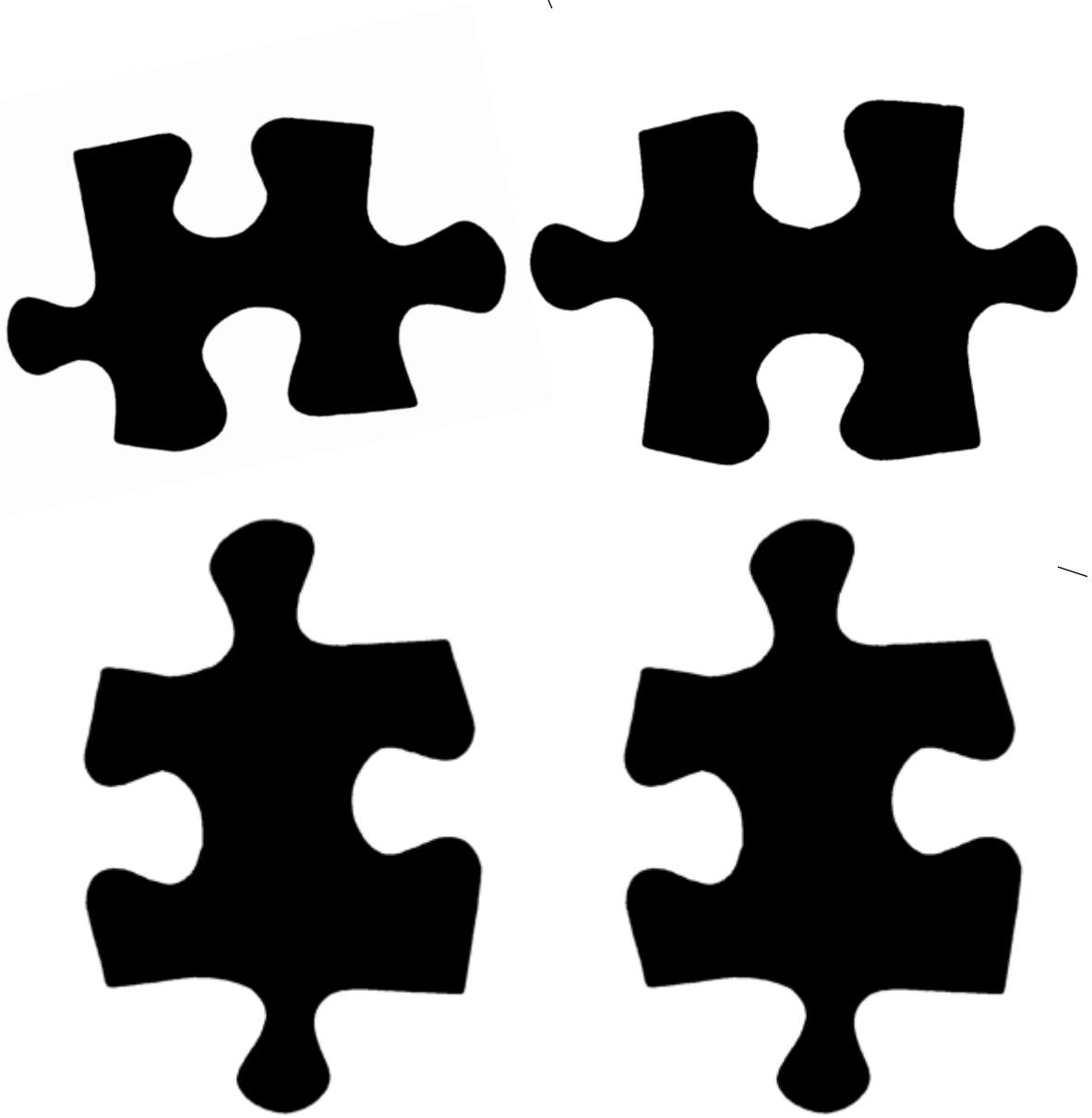
- Wir unterscheiden zwischen female, male und flat
- Kantentyp anhand von CCW Test



KANTEN TYP

- Wir unterscheiden zwischen female, male und flat
- Kantentyp anhand von CCW Test





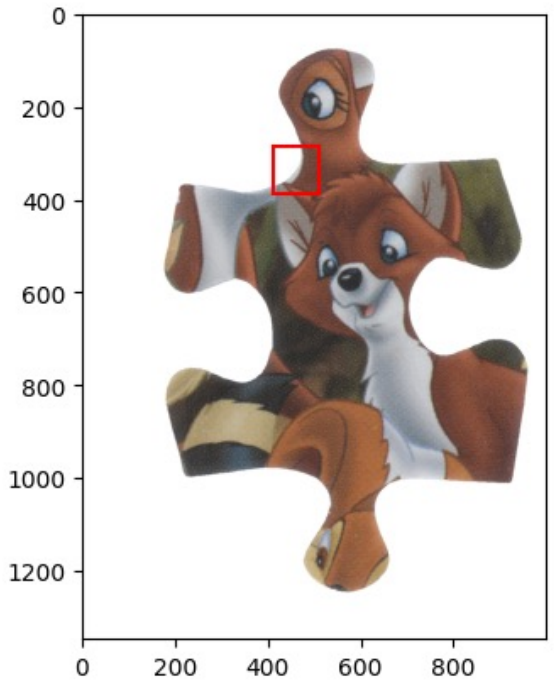
FARB MATCHING

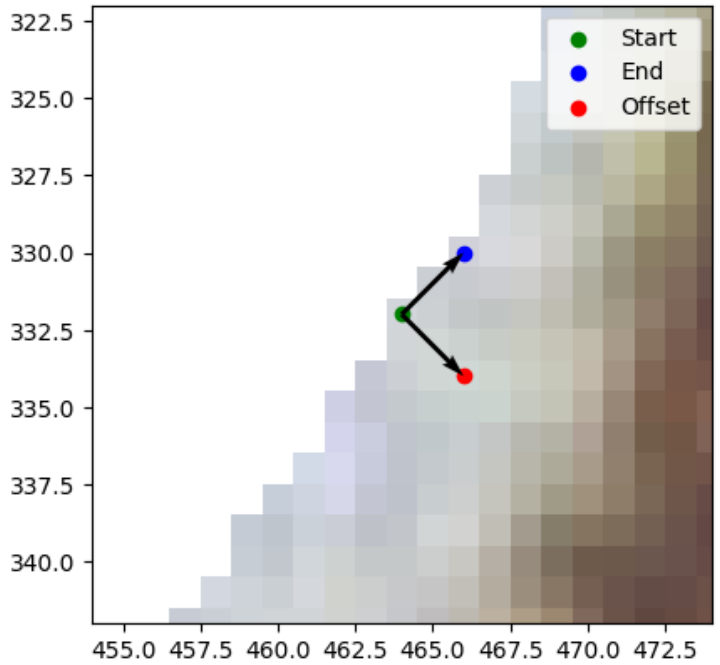


FARB MATCHING

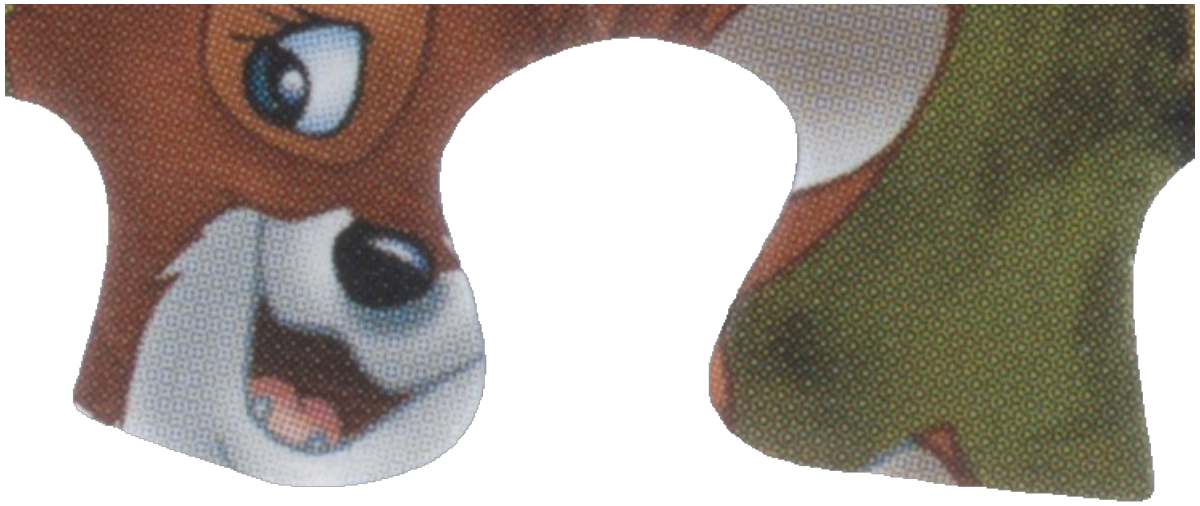
GEOMATCHER RGB

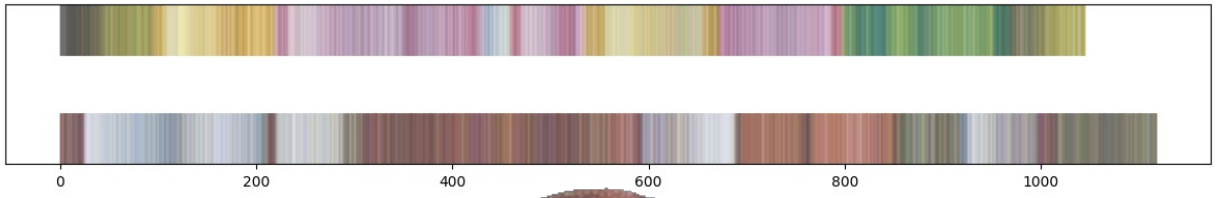
- Vorgehensweise wie beim Geomatcher aber statt Ortsinformationen (x,y) werden Farbwerte (r,g,b) extrahiert
- Summe der Unterschiede der Farbwerte ist das Gütekriterium zweier Kanten

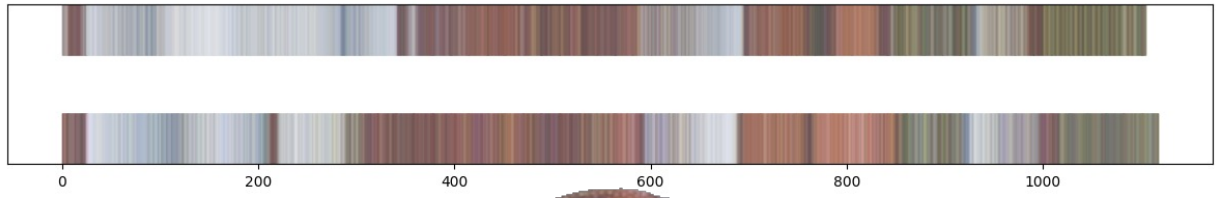




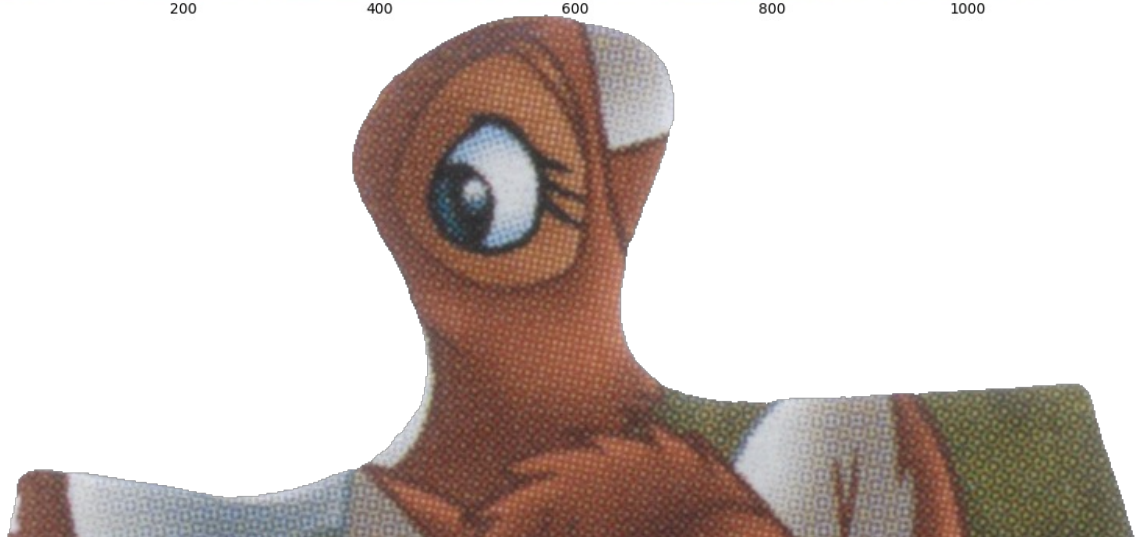


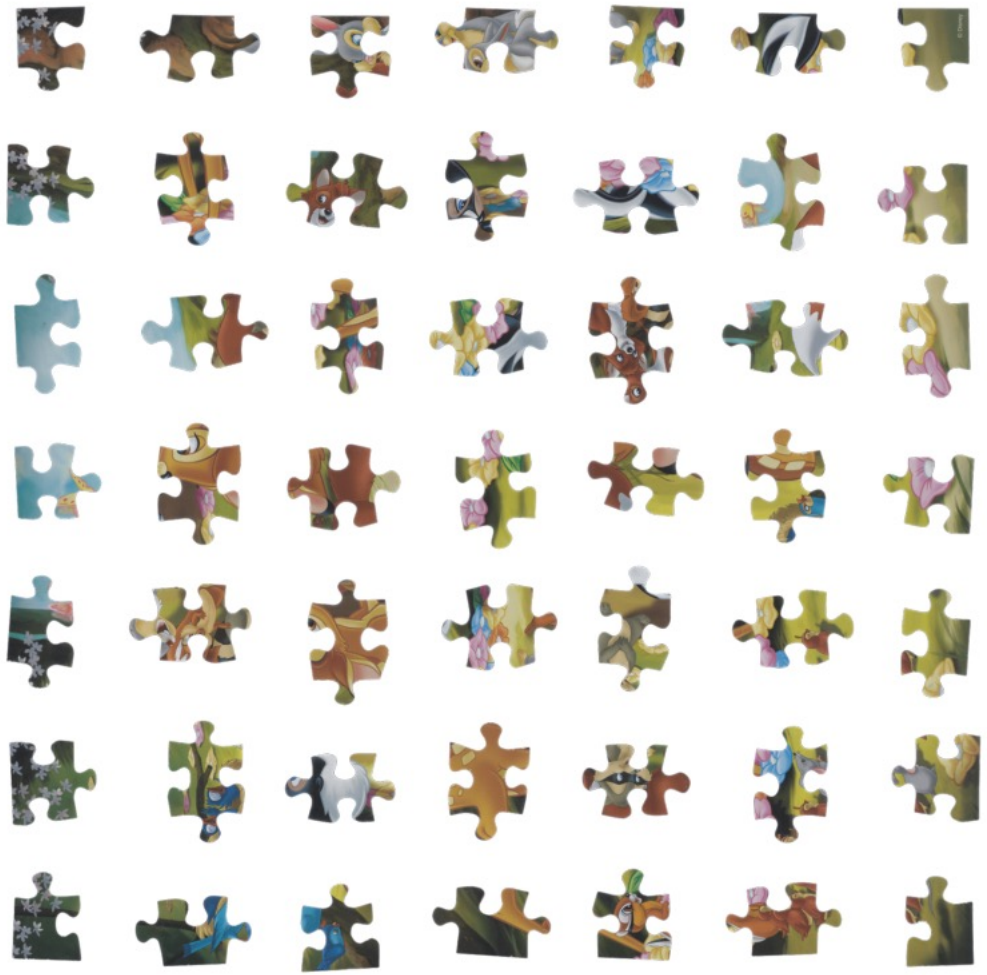



0
200
400
600
800
1000


0
200
400
600
800
1000







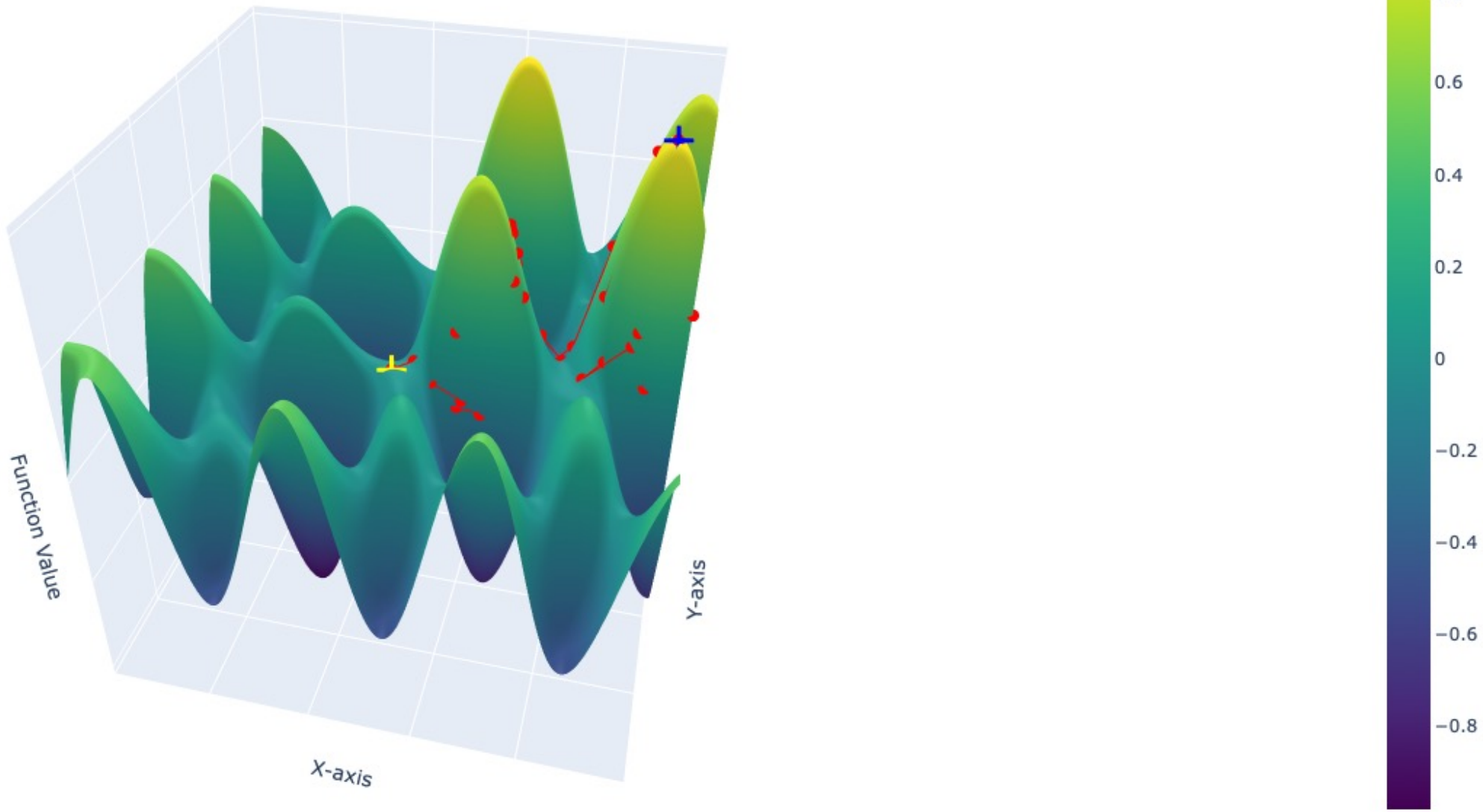
SIMULATED ANNEALING

PSEUDOCODE

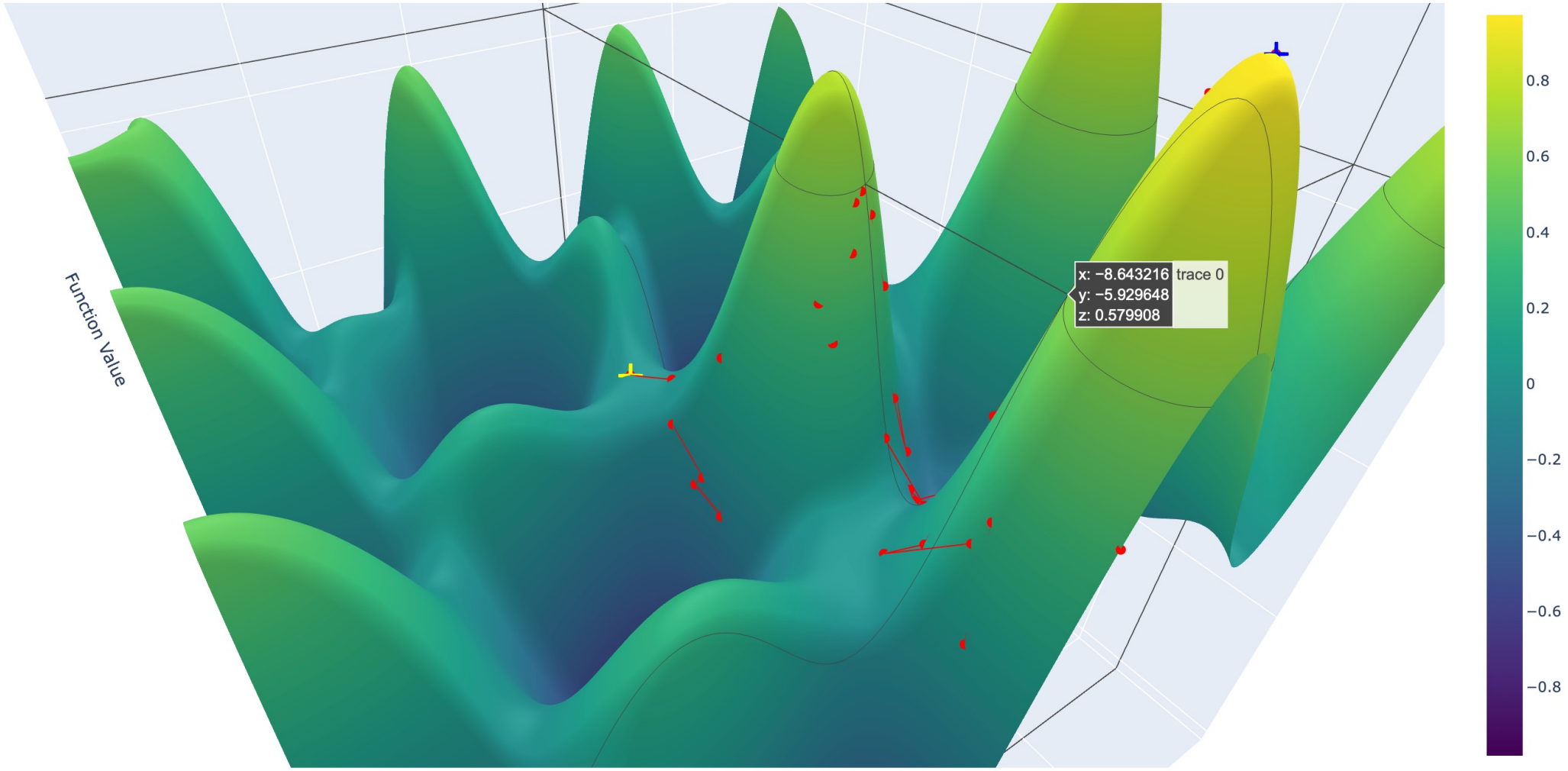
Algorithm 2 Simulated Annealing Algorithmus

Eingabe: Anfangszustand $\mathbf{s}_0$, maximale Iterationen $k_{\max}$, Anfangstemperatur T_0, Kühlrate α
Ausgabe: Finaler Zustand $\mathbf{s}$
$\mathbf{s} \leftarrow \mathbf{s}_0$
$T \leftarrow T_0$
for $k = 1$ bis $k_{\max}$ **do**
    $\mathbf{s}' \leftarrow \text{Bewegung}(\mathbf{s})$
    $\Delta E \leftarrow \text{Energie}(\mathbf{s}') - \text{Energie}(\mathbf{s})$
    if $\Delta E < 0$ **oder** $\text{zufall}(\Delta E, T)$ **then**
        $\mathbf{s} \leftarrow \mathbf{s}'$
    end if
    $T \leftarrow \text{AnpassungVonT}(T, \alpha, k)$
end for
return s

BEISPIEL

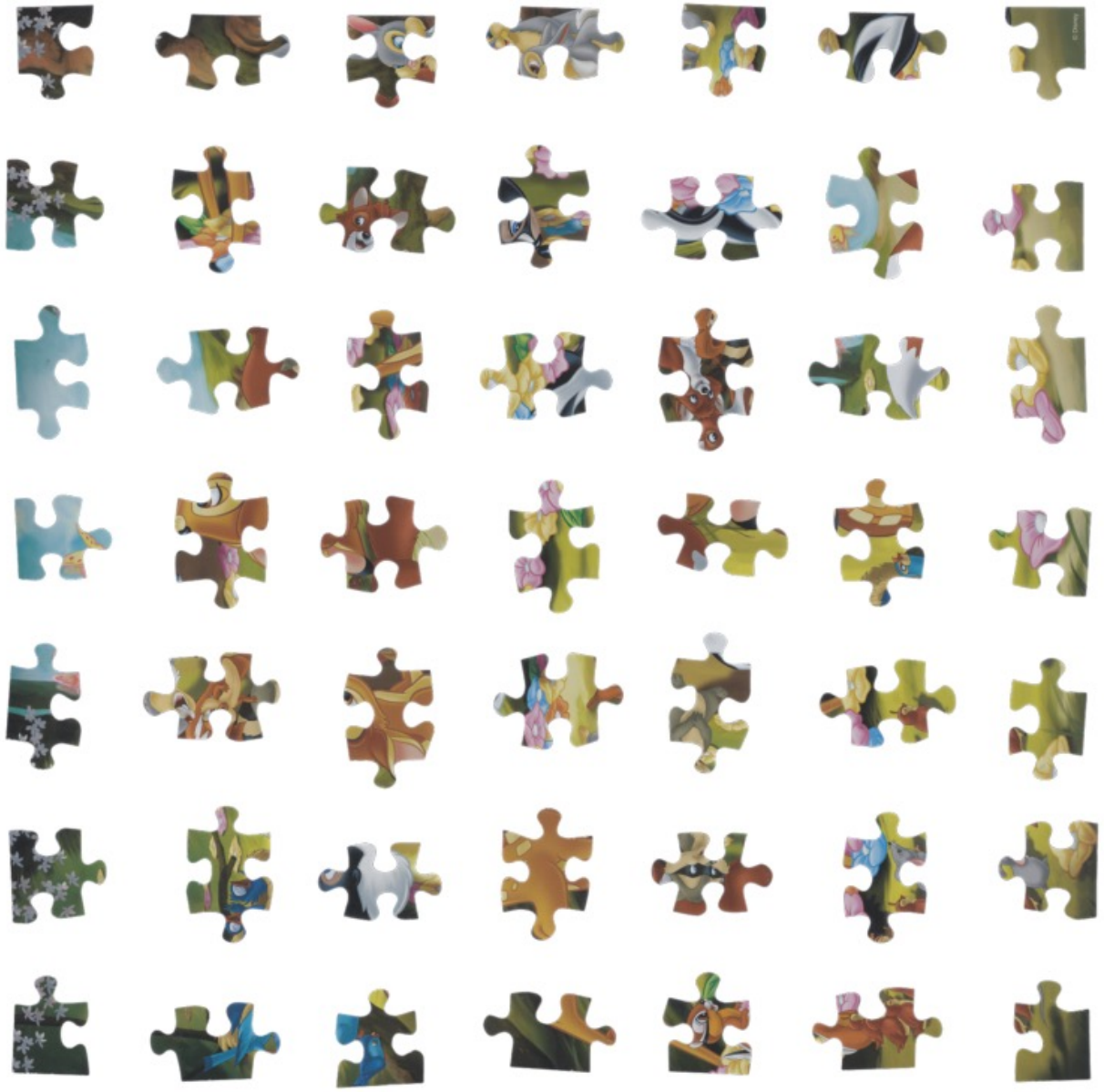


BEISPIEL


Function Value
x: −8.643216 trace 0
y: −5.929648
z: 0.579908
0.8
0.6
0.4
0.2
0
−0.2
−0.4
−0.6
−0.8

ERGEBNISSE

- Annealing nicht erfolgreich
- Zu viele Zustände
- Nicht effizient genug



VIELEN DANK FÜR IHRE AUFMERKSAMKEIT!

Simon Hampp

@simon-hampp (gitlab)